Analyzing Toronto Apartment Rental Prices

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Project Task Assignments

Austin did...

Olivia did...

Raina did...

Woobin did...

Introduction

In high-density metropolitan areas like Toronto, assessing how specific layout configurations influence rental pricing provides valuable insight into market structure and housing affordability. This study examines the relationship between apartment layout characteristics and monthly rent across the city. Specifically, we use multiple linear regression to evaluate how the number of bedrooms, bathrooms, and their interaction effects predict continuous rental prices. To test predictive validity further, we apply binary logistic regression using the same structural features to determine whether a listing classifies as a high-value property. By comparing main-effects and interaction models across both regression frameworks, we assess the overall explanatory power of layout features in Toronto's housing market.

Data Description

The dataset for this study was retrieved from a public Kaggle repository titled Toronto Apartment Rental Price, compiled by independent creator Raja CSP. According to the dataset documentation, the listings were aggregated and scraped from local rental websites active across Toronto in 2018.

The study utilizes Version 3 of the dataset, stored as a single 94.97 KB comma-separated values file. The raw file contains `n = 1,124` observational rows across 7 attribute columns, where each row corresponds to a unique apartment listing. There are no missing values in any columns in the raw data.

Exploratory Data Analysis

During data cleaning, we removed 308 duplicates which likely occurred because the data came from multiple sources. We also removed 9 rows where `price` was either below $100/month or above $6,000/month (namely, $65, $99, $99, $6,000, $6,500, $8,000, $9,750, $36,900, and $535,000) because these outliers may skew our results. The sample size is now `n = 807`.

Before modeling, we want to find out more about the data.

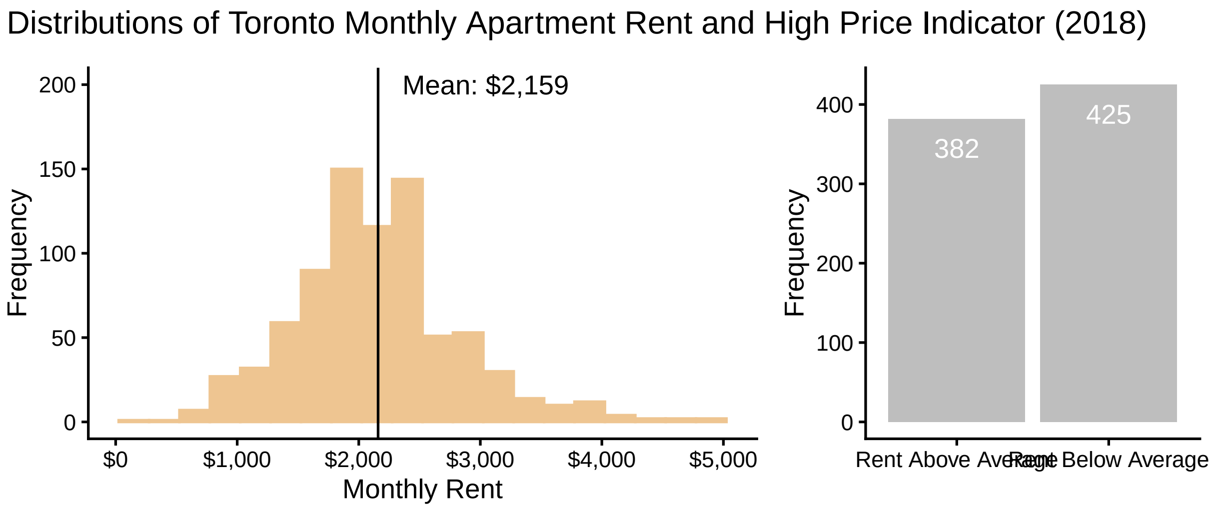
We first examine sample's summary statistics of each individual numerical variable.

Variable	Mean	Standard Deviation	Minimum	Maximum
bathroom	1.23	0.43	1.00	3.00
bedroom	1.36	0.56	1.00	3.00
den	0.15	0.36	0.00	1.00
lat	43.71	0.69	42.99	56.13
long	-79.50	1.85	-114.08	-73.58
price	2159.05	684.27	150.00	4900.00

The amount of `bathroom`s ranges from 1 to 3, the amount of `bedroom`s ranges from 1 to 3, and the amount of `den`s ranges from 0 to 1. The average amount of bedrooms is slightly higher than the average amount of bathrooms, and most apartment's don't have any dens, shown by its mean of 0.15. This makes sense - an apartment should have a bedroom and bathroom whereas a den is optional. Also, `latitude` and `longitude` fall in the valid range of [-90, 90] and [-180, 180] respectively (though we won't use these variables in modeling). So there's nothing unusual about these four variables.

As for `price`, its average value is $2,159/month with a standard deviation of $684/month. Its minimum is $150/month and its maximum is $4,900/month. While these minimum and maximum values are -2.59 and 4.01 standard deviations above and below the mean (respectively), we feel that they are reasonable enough to include in the scope of this analysis.

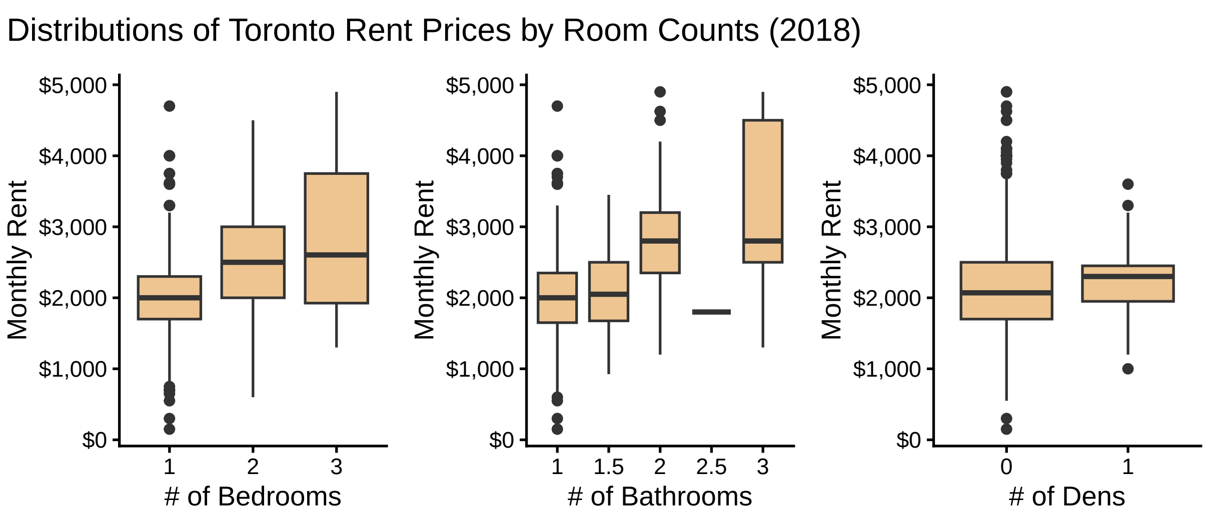
We now plot a histogram to visualize the distribution of `price` and a bar chart showing the class balance of the new binary variable `high_price`.



On the left, we see that `price` is approximately symmetric and unimodal around its mean. On the right, we see that the two `high_price` groups are approximately balanced; this means that logistic regression won't run into the issue of unbalanced response groups, which could bias modeling results.

Next, we examine possible relationships *between* variables.

Below are boxplots of `price` split by `bedroom`, `bathroom`, and `den` counts to see how `price`'s distribution varies by each of these apartment characteristics.



Looking independently at `bedroom`, `price`'s distribution slightly shifts upwards by a few hundred dollars for each additional bedroom. For `bathroom`, the distribution also moves upwards with the biggest drop from 2 to 2.5 bathrooms and another upwards shift in the distribution from 2.5 to 3 bathrooms. For `den`, the distribution is roughly equal for both groups with a difference-between-means of only $87/month. As well, within each group of boxplots, the distributions overlap with each other, so these predictors may not produce strong modeling results. Nonetheless, in terms of potential modeling power, `bathroom` may be better than `bedroom` and it may be better not to use `den`.

We're also curious about the correlations between the variables so we plot the upper triangle of a (Pearson) correlation matrix. Pearson correlation, denoted by the letter r, falls in [0, 1]. If two variables are positively correlated, it means that as one gets larger, the other also gets larger; if two variables are negatively correlated, it means that as one gets larger, the other gets smaller.

The variables with the strongest linear correlation with `price` are `bathroom` (r = 0.47) then `bedroom` (r = 0.39), though these are only moderately strong. In addition to the evidence from the boxplots, this further confirms that `den` is weakly correlated with `price` at r = 0.05. Moreoever, while `bedroom` and `bathroom` are moderately correlated at r = 0.63, putting models at risk of multicollinearity, both variables measure different characteristics of an apartment, so we may keep both during modeling.

In addition, here are frequency tables between `high_price` and `bedroom` and `bathroom`, excluding `den` because we've already seen that it would be a weak predictor.

# of Bedrooms	Rent Below Average	Rent Above Average	Total
1	62% (340)	38% (208)	100% (548)
2	32% (72)	68% (153)	100% (225)
3	38% (13)	62% (21)	100% (34)
Total	53% (425)	47% (382)	100% (807)

# of Bathrooms	Rent Below Average	Rent Above Average	Total
1	62% (381)	38% (232)	100% (613)
1.5	52% (12)	48% (11)	100% (23)
2	18% (30)	82% (135)	100% (165)
2.5	100% (1)	0% (0)	100% (1)
3	20% (1)	80% (4)	100% (5)
Total	53% (425)	47% (382)	100% (807)

Most of the data is on 1-bedroom or 1-bathroom units with the least data on units with 1.5, 2.5, or 3 bathrooms. Expectedly, most (62%) of the below-average units have 1 bedroom and 1 bathroom.

The main take-aways of this exploratory analysis are:

1. There is no class imbalance in `high_price` (for the purpose of logistic regression) because 53% vs. 47% of the observations have a `price` above vs. below average. Class imbalance, such as 90% vs. 5%, could bias a model's intercept and produce a poor model that maximizes accuracy by simply predicting the majority class.
2. In terms of predictive ability, `den` likely has the weakest linear relationship with `price`, as shown by the very similar boxplots and Pearson correlation of 0.05, and we exclude `latitude` and `longitude` because geographic effects on a target variable are probably not as linear or straightforward as these coordinates. This leaves `bedroom` and `bathroom` as the remaining covariates.

Models

Now that we have explored our data, we can consider models that can predict rental price based on our other variables. In the previous section, we narrowed down the variables we can use as predictors to number of bedrooms and number of bathrooms. To make use of these predictors, we have two options: a multiple linear regression and a multiple logistic regression.

Multiple Linear Regression

A multiple linear regression model model generates predictions as a straight line with a slope and intercept.

When making a multiple linear regression model, one of the first things to consider is whether or not interaction exists between covariates. If there is interaction, then the slope of the prediction line will be different depending on the value of the covariates; if not the slope is constant. This can be checked with hypothesis testing, something which can be checked by creating a model with interaction in R and checking the model's statistics.

Results with interaction:

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	1496.02	210.86	7.09	0.00	1082.12	1909.93
bedroom	0.95	125.45	0.01	0.99	-245.30	247.21
bathroom	318.44	180.91	1.76	0.08	-36.68	673.56
bedroom:bathroom	146.84	90.94	1.61	0.11	-31.67	325.35

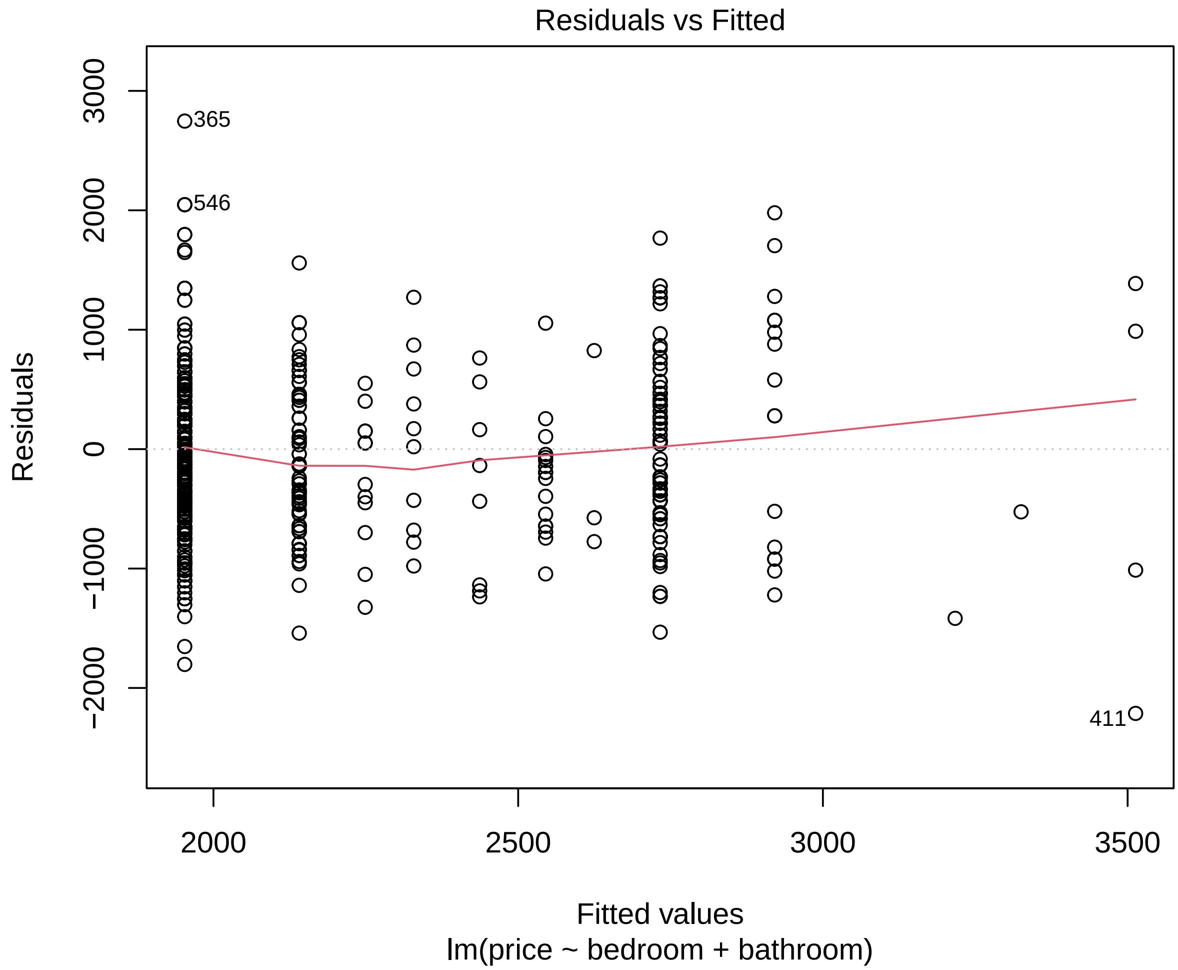
Results without interaction:

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	1172.76	66.25	17.70	0	1042.72	1302.80
bedroom	187.90	48.35	3.89	0	93.00	282.80
bathroom	592.21	63.16	9.38	0	468.23	716.19

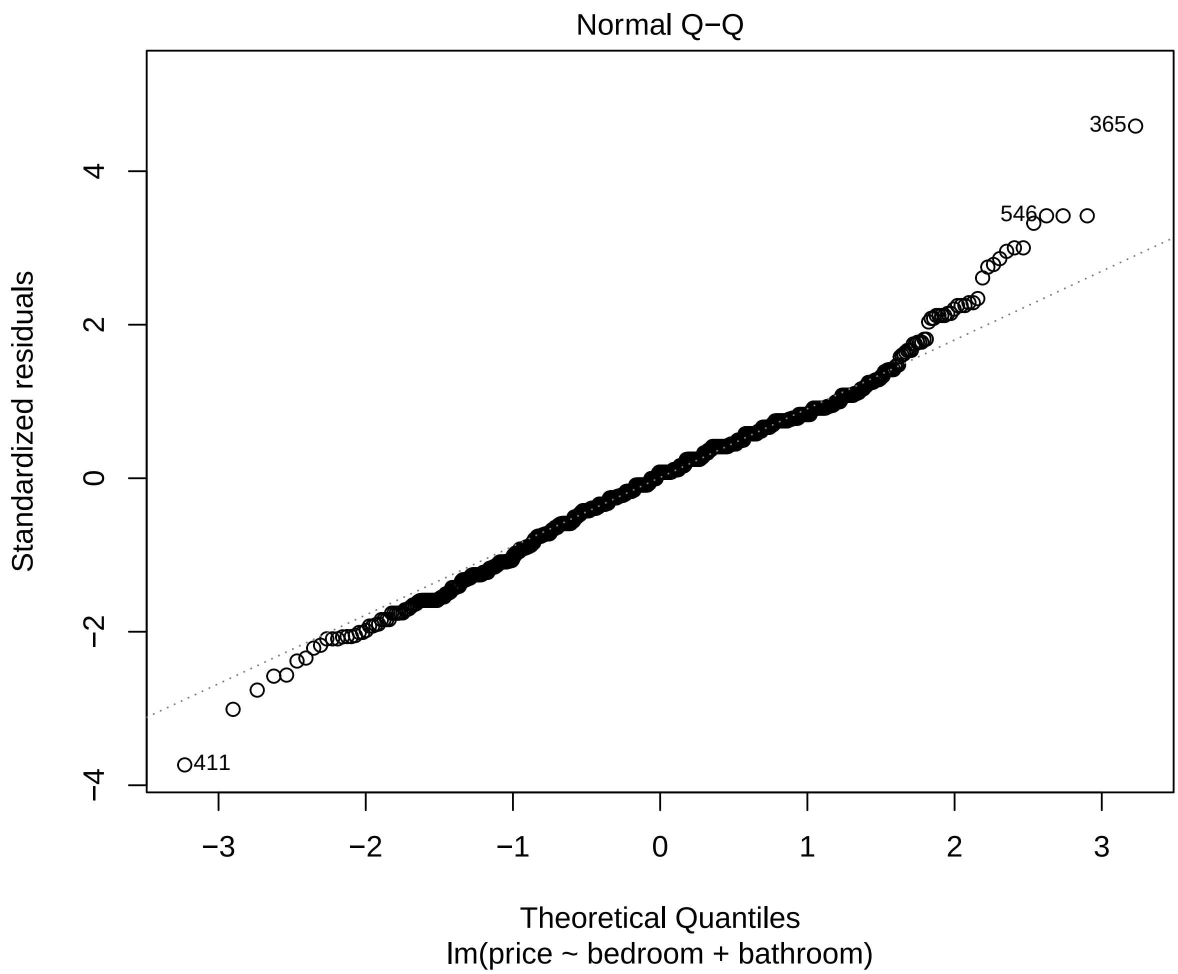
As shown in the tables above, the table produced by the model with interaction is reporting p-values in excess of our predecided level of significance (95%, necessitating a value below 0.05). Thus, we cannot confidently identify any interaction between the predictors and must proceed with the model without interaction.

The estimate column in the table for the non-interaction model shows that our prediction line will have an intercept of 1026.44. It shows that an additional bathroom in a property is associated with a $206.60 increase in monthly rent when number of bathrooms is held constant and that an additional bathroom in a property is associated with a $707.66 increase in rent when number of bedrooms is held constant.

To ensure that our linear model is appropriate for the data, we need to run some model diagnostics in the form of a residual and Q-Q plot.



For a linear model to be appropriate, the error around the trend/prediction line should be consistent and normally distributed. Residual plots offer a convenient visual representation of this error. In a residual plot, error should be distributed normally and evenly around a horizontal line at y = 0. As you can see in the plot above, because our predictors are categorical, the data points are not continuously distributed, making the plot somewhat messy to look at. It shows that while the error approximately follows a normal distribution distribution, but with larger variance around certain areas in the x axis, meaning the variance is not perfectly consistent.



Another way of checking the distributon of the error is a Q-Q plot. If the error is normally distributed with equal variance, then a Q-Q plot will show all data points distributed along a straight, sloped line wherein X = Y. The Q-Q plot of our linear model shows the data following this line for the most part, but diverging at the beginning and the end, implying that our model has more extreme data than can be expected under a true normal distribution (Clay, 2015). Applying transformations to the model, such as log transformation and predicting the square root of the response variable, were unsuccesful in creating a more favourable distribution.

While our diagnostics reveal that our data may not be perfectly suited for a linear analysis, this is not uncommon for observed data. It is our belief that our model is still useful for making predictions.

Multiple Logistic Regression

A multiple logistic regression model is used to make predictions for a categorical response variable. To make use of such a model, we converted our response variable (price) into a categorical one by splitting it into two categories for rent above and below the mean value. Thus, our model will be predicting whether a property has above or below average rent, rather than a specific value.

Once again, we need to begin by checking for interaction between predictors.

Results with interaction:

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	-0.30	0.17	-1.82	0.07	-0.62	0.02
bedroom	0.20	0.10	2.05	0.04	0.01	0.39
bathroom	0.58	0.14	4.13	0.00	0.31	0.86
bedroom:bathroom	-0.12	0.07	-1.70	0.09	-0.26	0.02

Results without interaction:

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	-0.04	0.05	-0.68	0.50	-0.14	0.07
bedroom	0.05	0.04	1.26	0.21	-0.03	0.12
bathroom	0.36	0.05	7.27	0.00	0.26	0.46

The tables above show that neither model passes the hypothesis tests necessary to prove statistical significance. Of the two models, the results with interaction come much closer. To proceed with our analysis, our only option is to lower our stated level of significance to 89% and continue with the model with interaction. So, keep in mind that any further observations on our logistic regression model will be made with this altered level of significance.

The table of our model with interaction shows that an additional bedroom increases the log odds of a property having high rent by 0.19, an additional bathroom increases the log odds of a property having high rent by 0.57, and…

Again we must run some model diagnostics to ensure our model is appropriate for the data. This time we will use a binned residuals plot.

(insert binned residuals plot here).

In a binned residuals plot, the majority of data points (approximately 95%) must fall within the generated error bounds (Ludecke et al., n.d.). As you can see, our model easily meets this criteria as all of the data points fall within these bounds.

Conclusion

The multiple linear regression analysis indicates that both the number of bedrooms and the bathrooms have a positive relationship with the monthly rent of a property. The multiple logistic regression analysis indicates that both the number of bedrooms and the bathrooms have a positive relationship with the monthly rent. However, as we can see from the residual plot and the Q-Q plot, the key assumptions of homoscedasticity and normality of residuals are invalid. These violations affected to the prediction confidence intervals and the p-values. Namely, the current MLR model results cannot be considered fully reliable.

In the logistic regression model predicting whether rent is above average, applying an adjusted 89% confidence level yields a statistically significant positive effect for the number of bathrooms on the likelihood of having above-average rent. This founding relies on an adhoc modification of the significance threshold to fit the observed data. Consequently, the findings from the logistic regression model should be interpreted with caution.

To improve the model's reliability, three key adjustments are recommended for future research:

Re-introducing the dropped location coordinates will help reducing unexplained variance and standard errors,thereby addressing potential omitted variable bias. Applying a log transformation to the response variable (ln(price)) along with removing extreme outliers will normalize the right-skewed price distribution, helping to satisfy the linear regression assumptions of normality and homoscedasticity. Lastly, setting the significant level to the standard 95% confidence threshold will ensure the hypothesis testing remains statistically unbiased.

References

References will go here.

Appendix

```r
# Loading packages
library(tidyverse)
library(knitr)
library(scales)
library(patchwork)
library(corrplot)
library(broom)
```